Sammy Al Hashemi

Device Characterization Engineer

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(647)-463-2638

cGPA: 3.56

Hardware Skills

Verilog

80%

Electronic Circuit Analysis

80%

Software Skills

Python

80%

Angular

75%

nodeJS

60%

Courses Taken

- Algorithms: ECE358
- Data Strcutures & Algorithms: CSC190
- Machine Learning and Data Mining: CSC411
- Electronic Circuits: ECE354
- Digital Electronics: ECE334

Website

www.sammyalhashemi.com

Work Experience

Device Characterization and Product Engineering Intern

Microsemi Corp.

July 2017 - August 2018

- Worked on development of test structures and designs using Microsemi's design software, Libero, to assess and characterize Microsemi FPGA products that went to market.
- Specifically, characterized the Mathblock component of Microsemi's G5 FPGA product as well as performance-related AC tests on other combinatorial and sequential elements found in the FPGA.
    - Example components include look-up tables, carry chains, digital I/O's.
    - Mainly characterized in terms of maximum frequency.
    - Tested chains on different corners of the FPGA fabric.
    - AC characterization was done on Verigy 93k tester offsite using test vectors generated on site.
- Inside Libero, I created Verilog test structures, making use of JTAG to read input from test programs created using a company code.
    - Also used Libero-specific SmartTime software to enforce timing constraints in my designs and check for any violations.
- Created testbenches and wave.do files to perform simulations of the test design using the ModelSim environment.
    - The wave.do files mimicked the signals normally fed into the system by the test programs.
- Additionally assisted with the Product Engineering team in programming devices pre and post burn-in.
- Wrote sections of the characterization report found in the official product manual.

Thesis

July 2017 - Present

- Today's classical computers do not carry the capacity to be able to break modern encryption algorithms without first knowing secret keys that initialize the algorithm.
- Using IBM Q's python SDK, I will be programming scalable versions of Shor's and Grover's algorithms for my thesis under Proffessor Glen Gulak.

Education

University of Toronto

Bachelor of Applied Science and Engineering *Department of Engineering Science*

Sept 2014 - Present

Enrolled in the Engineering Science Program. I chose the Physics option which focuses on engineering design with major physics applications. However, it also granted great flexibility for courses tailored towards my main interests of tech and quantum mechanics.

Personal Projects

Personal Website

- Single-page application written in an Angular6 frontend with Typescript
- Uses Firebase's Cloud Storage to store larger files such as pictures.
    - Routes with photos that are stored in the cloud have their URLs pre-fetched using a service implementing Angular's `Resolve<>` interface.
    - The URLs are loaded asynchronously using Angular's `async` pipe.
- Animations used to liven up pages include:
    - Loading indicator when pre-fetching data
    - Route animations and child animations upon page load
- Uses complex angular routing features
    - data transfer with route parameters
    - asynchronous picture loading with Firebase Storage
    - route resolvers
    - lazy loading
- RxJS usage for asynchronous data handling
    - During development, I also migrated from RxJS5 to RxJS6 to keep up-to-date.

Tennis Scraper

- Writing a webapp (still in development) that webscrapes the ranking history of tennis players Roger Federer, Rafael Nadal, Novak Djokovic, Andy Murray, and Nick Kyrgios.
- The ranking data is displayed on a chart vs. year they played.
- Front end is written in react. Main class components are:
    - Top level app component that passes the chosen player's name down.
    - Body component that calls the backend webscraper, passing the result down.
    - Graph component that recieves the result of the webscraper and plots the data using `chartist.js`.
- Backend webscraper is an `express` server that uses `cheerio.js` wrapped in a request-promise.
    - `Cheerio.js` scrapes the player's ranking page by element selector.

"Restaurant Finder"

- Wrote an iOS app in Swift3 in Xcode.
- Personalized list of branches where I could look up the closest locations to me.
- App structured using MVC.
- Used Alamofire to access the Google Maps/PlacesAPI.
- De-structured the returned JSON to supply the user with direction information.
- This was an introductory app that really wasn't that great at all, but I'm proud of the fact that it was my first.